Ships and Offshore Structures

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/tsos20

The bearing capacity of A-shaped mat foundations on cohesive soil

Run Liu, Runbo Cai, Guangsi Chen & Chao Liang

SHIPS AND OFFSHORE STRUCTURES
https://doi.org/10.1080/17445302.2022.2116208

The bearing capacity of A-shaped mat foundations on cohesive soil

Run Liu, Runbo Cai, Guangsi Chen and Chao Liang

State Key Laboratory of Hydraulic Engineering Simulation and Safety, Tianjin University, Tianjin, People's Republic of China

ABSTRACT

Because of its irregular shape, it is inaccurate to calculate the bearing capacity of the A-shaped mat foundation by a conventional method, which may have potential safety hazards. In this paper, the vertical bearing capacity of the mat is obtained by using the equivalent calculation method (ECM) and the finite element method respectively, and compared with the centrifuge test results. A correction method based on the ECM is proposed. Influences of foundation dimensional parameters on the *H*, *M* and *H-M* combined bearing capacity are studied. Expressions of *H-M* failure envelopes related to the dimensional parameters are established. A new form of *V-H-M* envelope expression for mats is suggested according to the effect of vertical load on *H-M* capacity. The modified ECM and the *V-H-M* envelope proposed in the paper can be used to quickly evaluate the vertical bearing capacity of the A-shaped mat and its stability under coupling loads.

ARTICLE HISTORY
Received 25 April 2022
Accepted 17 August 2022

KEYWORDS
Centrifuge test; equivalent calculation method; dimensional parameters; failure envelope

1. Introduction

Mat foundations are usually used for offshore mobile platforms. The platform sits on the seabed through a large mat connecting each leg at its bottom. Mat foundations have been widely used in offshore structures (e.g. jack-up rigs and installation platforms) with the development and utilisation of natural resources such as wind energy, offshore oil and gas all over the world. The self-weight load of the offshore platform is quite considerable because of its large size and facilities on it. The large bottom area of the mat reduces the pressure of the platform acting on subsoil. This greatly avoids punch-through accidents and is conducive to improving platform stability (Hirst et al. 1976; Boswell L et al. 2017; Li et al. 2021). It can also act as a damper to weaken the heave motion of the platform (Parikh et al. 2021). The hollow mat is usually divided into compartments. The buoyancy can be increased by drainage to offset part of the vertical load if the upper load is too large. While the buoyancy can be reduced by injecting water in case of extreme weather. Make the mat sink slightly, increase the embedded depth and foundation pressure, and resist wind and wave loads. At the same time, the compartment can be designed as an oil storage tank to improve the efficiency of the oilfield production and transportation system (Fan 2014). The huge flat hollow box makes mat foundations have many advantages and broad application prospects. It is noteworthy that an accurate prediction of the bearing capacity is the premise and key factor of the application of mats.

The mat is a kind of offshore shallow foundation. Early research on mats mainly refers to the results of onshore shallow foundations. The shallow foundation bearing capacity subject began with the research on two-dimensional strip foundations by scholars such as Prandtl (1921), Davis E and Booker J (1973). After that, it continues to develop to three-dimensional foundations such as Meyerhof (1953), Hansen (1970) and other scholars' research on foundation shape, load angle, load eccentricity and other issues. Based on this, a series of design guidelines such as DNV (1992) and API (2014) were formed and have been widely recognised and applied by scholars and engineers around the world. In recent years, research on the uniaxial bearing characteristics of shallow foundations has been further developed. Chen and Liu (2018) applied the upper bound theorem to study the vertical bearing capacity of mudmat foundations on sand, and the concept of soil damage rate was proposed. Liu et al. (2019, 2020) analyzed the effects of perforation ratio, perforation number, and heterogeneity factor on the vertical bearing capacity of rectangular foundations on clay by combining centrifuge test and finite element method (FEM). In addition, the loads on offshore structures come from environmental factors such as wind, wave and current. The load direction is complex and changeable, so it is unsafe to analyze the uniaxial bearing capacity only. The bearing characteristics of the foundation under multi-dimensional load should be comprehensively considered. Taiebat H and Carter J (2000) studied the failure envelope of circle foundations under multi-dimensional load on cohesive soil by establishing the three-dimensional finite element (FE) model. Gourvenec (2007) analyzed the shape effect of undrained bearing capacity of rectangular foundations on heterogeneous soil under *V-H-M* load. Feng et al. (2017) further obtained the relationship between bearing performance under *V-H-M* load and soil heterogeneity factor and foundation aspect ratio. Shen et al. (2017) studied the interface conditions between subsoil and mudmat foundations, and established envelope expressions. Liu et al. (2021) considered torque load in addition to *V-H-M* load, and assessed the bearing characteristics of square foundations under *V-H-T* load.

A boat-shaped bow is usually designed in the mat foundation to reduce the towing resistance of platform, so the mat's top view is approximately A-shaped (as shown in Figure 1). The bow of mat usually has a perforation to reduce the foundation weight and the uplift force. A wellhead groove is set at the stern of mat applied to a jack-up rig for the drilling operation. In the middle of the mat, a cross-member connects the left- and right-side parts to provide sufficient stiffness.

The irregular shape of the A-shaped mat may cause its bearing characteristics to be different from the rectangular foundation, square foundation and other regular shaped foundations. It is necessary to analyze the bearing capacity of mats in multi-dimensional load space. Therefore, this paper comprehensively applies

CONTACT Chao Liang liangchao@tju.edu.cn State Key Laboratory of Hydraulic Engineering Simulation and Safety, Tianjin University, Tianjin 300072, People's Republic of China

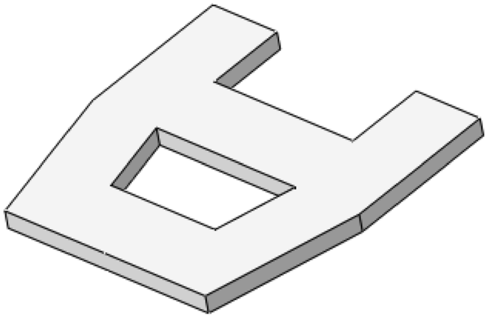


Figure 1. Schematic diagram of the A-shaped mat foundation. (This figure is available in colour online.)

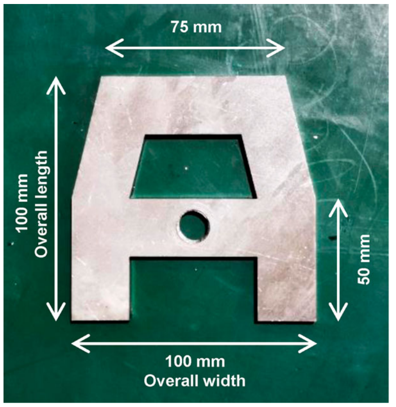


Figure 3. A-shaped mat test model. (This figure is available in colour online.)

the centrifuge test, equivalent calculation method (ECM) and FEM to assess the vertical bearing capacity of mats on cohesive soil, and the correction method of ECM is proposed. Influences of foundation dimensional parameters on bearing characteristics under uniaxial and multi-dimensional load conditions are studied. A new form of three-dimensional *V-H-M* failure envelope expression which is more in line with the bearing capacity law of A-shaped mats is constructed. The modified ECM and the *V-H-M* envelope proposed in the paper can be used to quickly evaluate the vertical bearing capacity of the A-shaped mat and its stability under coupling loads.

2. Centrifuge test

2.1. Experimental set up

The centrifuge test was carried out to accurately evaluate the vertical bearing capacity of A-shaped mat foundations and to provide a reference for subsequent FE analysis. The TLJ-100A centrifuge located at the Institute of Geotechnical Engineering of Tianjin University has a capacity of 100 g·t, an effective radius of 2.7 m, and a maximum overload acceleration of 200 g, as shown in Figure 2.

Kaolin clay was used in the test. It was measured to have a plastic limit of 28.2% and a liquid limit of 58.1%. Clay powder was fully mixed with water according to twice the liquid limit and poured into the strongbox where the dry sand drainage layer and geotextile were laid, and then the geotextile and dry sand were laid again on the surface of clay. The internal dimensions of the strongbox used in the test are 880mm × 595mm × 400 mm (length×width×height). The clay went through a backpressure consolidation process, and the soil strength was measured regularly. After consolidating to the target strength, the stiff top layer soil was removed and the soil surface was flattened. A 20 mm layer of water was maintained to keep the saturation.

Figure 2. The TLJ-100A centrifuge. (This figure is available in colour online.)

It can be determined that the typical mat length is about 50 m referring to the existing mat foundations design and engineering application examples (Hirst et al. 1976; Young A et al. 1982; Zhang and Wang 2015). It is difficult to achieve 50 m prototype foundation test considering the size of the strongbox and the load capacity of the centrifuge. Thus the model foundation with the overall dimension of 100mm × 100 mm (length×width) was processed, as shown in Figure 3. The similarity scale was 1:100, i.e. the prototype foundation length was 10 m. The test data in the following text are expressed in the prototype scale unless otherwise specified.

2.2. Experimental programme

In order to prevent the tilting problem in the test, the model mat and the loading rod were made of stainless steel with high stiffness, and the mat and the loading rod were connected at the centre of cross-member close to the centroid of the foundation. They were connected by the thread running through the foundation. A load cell and a hydraulic cylinder were connected above the loading rod, as shown in Figure 4. The test started after connecting devices correctly. The acceleration of the centrifuge was raised gradually from 1 to 100 g, and it was maintained for a period of time to make soil stress field stable. The mat was lowered by the hydraulic cylinder so that its bottom was just in contact with the soil surface. The mat was completely submerged, and the sensor value was zeroed to eliminate the influence of buoyancy on the test results. A

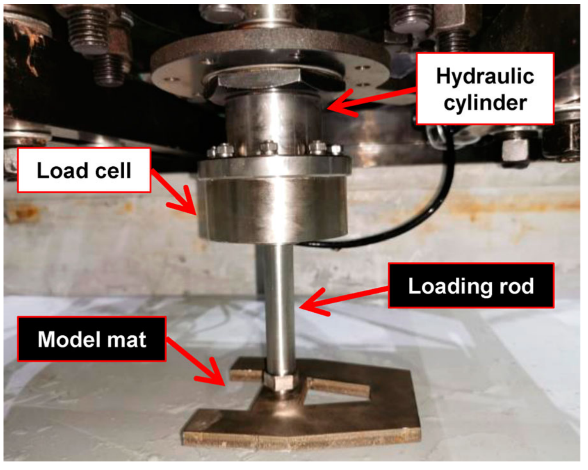


Figure 4. Test loading devices. (This figure is available in colour online.)

Figure 5. Centrifuge test process. (This figure is available in colour online.)

uniform loading velocity was used to slowly press down the mat to the specified depth. The vertical displacement of the mat and the soil reaction force were measured during the test.

2.3. Test results

The centrifuge test process of vertical bearing capacity centrifuge of mat foundation is shown in Figure 5.

As can be seen from Figure 5, the clay was squeezed and refluxed from the perforation and the edges of mat, covering the top of mat gradually with the increase of penetration depth. Finally, the foundation made soil surface sunken slightly. This is due to the high acceleration condition of the centrifuge, resulting in a high level of soil stress. It is difficult to form a bulge with significant self-supporting height for the clay squeezed by the mat. The soil constantly flowed back to the foundation and formed a depression.

The vertical load-displacement curve of the mat and the strength distribution of the soil in the centrifuge test are shown in Figures 6 and 7.

It can be seen from Figure 6 that the vertical load still increases gradually when the mat penetrated into 3 m, but it tends to be flattened. The intersection point of two tangent lines at the beginning and end of the curve acts as the ultimate vertical load with a value of 2522 kN, according to the method used by Graham et al. (1982). It can be seen from Figure 7 that the strength of the test soil is relatively uniform, while it is slightly smaller only at the soil surface. The surface of clay was easy to be disturbed due to the presence of overlying water, which may cause small measuring results. It can be approximately regarded as homogeneous clay with strength of 6 kPa, considering the penetration depth of the foundation and a certain influence depth of the vertical bearing capacity.

3. Vertical bearing capacity

3.1. Equivalent calculation method

The API recommended practice (2014) states that a shallow foundation with irregular shape can be idealised to a rectangular foundation to calculate the bearing capacity, making the idealised shape and the real shape:

(1) Having the same area

$$A_{\text{idealised}} = BL = A_{\text{real}} \tag{1}$$

(2) Having the same areal moments of inertia

$$I_{x,\text{idealised}} = I_{x,\text{real}} = \int_A y^2 dA \tag{2}$$

$$I_{y,\text{idealised}} = I_{y,\text{real}} = \int_A x^2 dA \tag{3}$$

where: $A_{\text{idealised}}$ is the idealised foundation area; A_{real} is the original foundation area; B and L are width and length of the

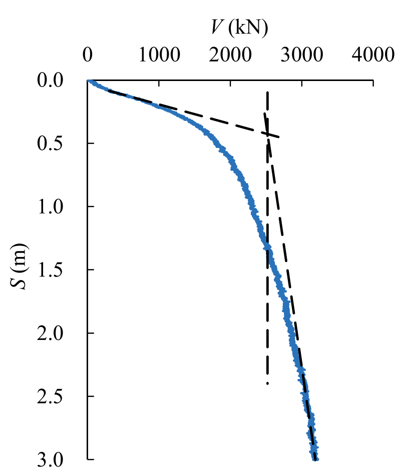


Figure 6. Vertical load-displacement curve (test). (This figure is available in colour online.)

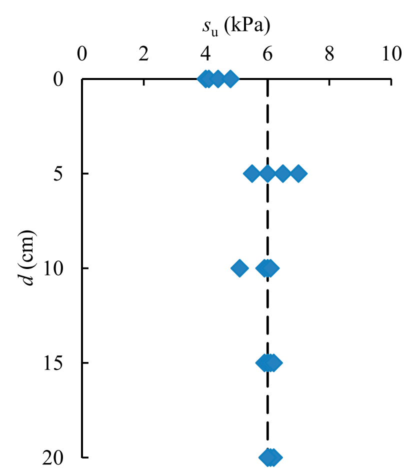


Figure 7. Soil strength distribution. (This figure is available in colour online.)

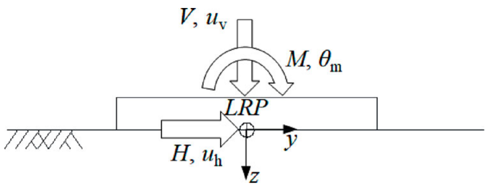


Figure 8. Loading condition for the mat. (This figure is available in colour online.)

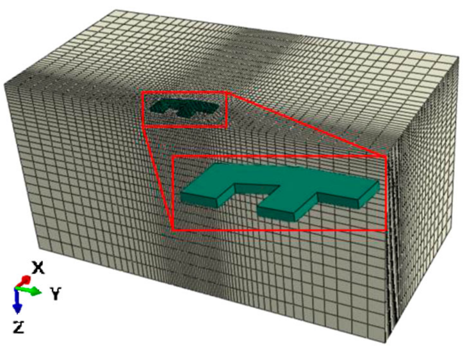


Figure 9. FE model of the mat and soil (half view). (This figure is available in colour online.)

idealised rectangular foundation, respectively; $I_{x,\ \text{idealised}}$ and $I_{y,\ \text{idealised}}$ are the idealised foundation's areal moments of inertia about x-axis and y-axis, respectively; $I_{x,\ \text{real}}$ and $I_{y,\ \text{real}}$ are the original foundation's areal moments of inertia about x-axis and y-axis, respectively.

Equations (1)–(3) generally cannot be satisfied at the same time for the subject that equate an arbitrary shape to a rectangle, so the issue is divided into two directions: For action effects in the y-direction, Equations (1) and (2) should be satisfied. For action effects in the x-direction, Equations (1) and (3) should be satisfied.

The ultimate vertical load V_{ult} of surface rectangular foundation under undrained conditions can be calculated according to Equation (4) (API RP 2GEO 2014).

$$\begin{cases} V_{\text{ult}} = q_{\text{u}} B L \\ q_{\text{u}} = s_{\text{u}} N_{\text{c}} K_{\text{c}} \\ K_{\text{c}} = 1 + s_{\text{c}} \\ s_{\text{c}} = 0.18B/L \end{cases} \tag{4}$$

where: q_{u} is the vertical bearing capacity; s_{u} is the undrained shear strength of soil; N_{c} is the bearing capacity dimensionless constant (equal to 5.14); K_{c} is the correction factor; s_{c} is the shape factor.

3.2. Finite element method

The FE software ABAQUS (Dassault Systèmes 2011) is used to analyze the bearing capacity of the A-shaped mats. The load direction is set based on the method suggested by Butterfield et al. (1997). A load reference point (LRP) is set at the centroid of the interface of the mat and soil, as shown in Figure 8. The LRP is coupled to the foundation surface to control the three-dimensional motion of the mat.

Table 1. Geometric parameters.

Action effects	Foundation basal area (m^2)	I_x (m^4)	I_y (m^4)	B_x' (m)	B_y' (m)
x-direction	68.40	/	650.21	10.68	6.40
y-direction	68.40	534.66	/	7.06	9.69

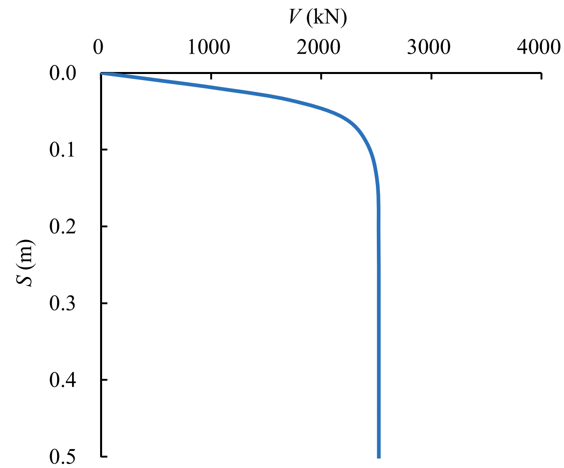


Figure 10. Vertical load-displacement curve (FEM). (This figure is available in colour online.)

The undrained soil condition is represented a linear elastic perfectly plastic constitutive based on the Tresca yield criterion. The undrained shear strength s_{u} is set to 6 kPa and the buoyant unit weight γ' is 6 kN/m^3 according to the test results. Take Poisson's ratio ν to 0.49 to simulate undrained conditions. Young's modulus E of the soil has little effect on the bearing capacity, so it is taken as $500s_{\text{u}}$ to reduce the calculation cost and avoid mesh distortion as much as possible.

The stiffness of the mat is significantly greater than that of the soil, so the foundation is set as a rigid body. The bottom of the mat is located at mud line. The foundation-soil interface is set to unlimited tension condition as recommended by Liu et al. (2021).

After the sensitivity analysis, the soil is modelled according to the following rules considering the calculation accuracy and cost: The mesh extends $7B_{\text{x}}$ (B_{x} is the length of the mat along the x-axis) in the horizontal direction and $3.5B_{\text{x}}$ in the depth direction. The minimum mesh size is $0.04B_{\text{x}}$ (near the mat) to simulate failure mechanisms accurately, and the maximum mesh size is $0.3B_{\text{x}}$. Horizontal displacement constraint is applied at the vertical boundary, and full displacement constraint is applied at the bottom boundary. A typical FE model is shown in Figure 9 and it comprises 71,000 elements approximately.

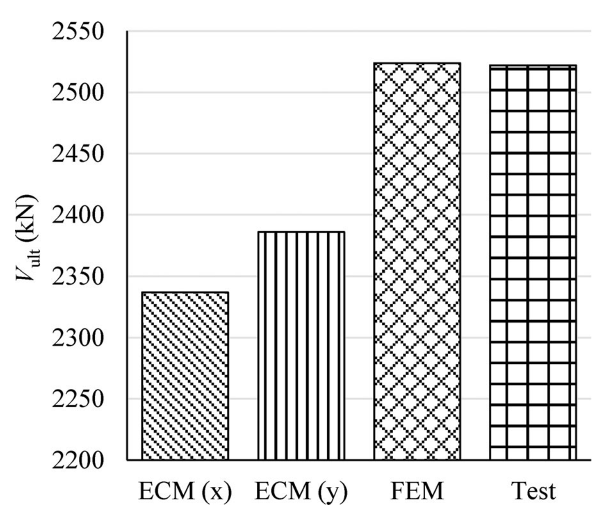


Figure 11. Comparison of vertical bearing capacity results. (This figure is available in colour online.)

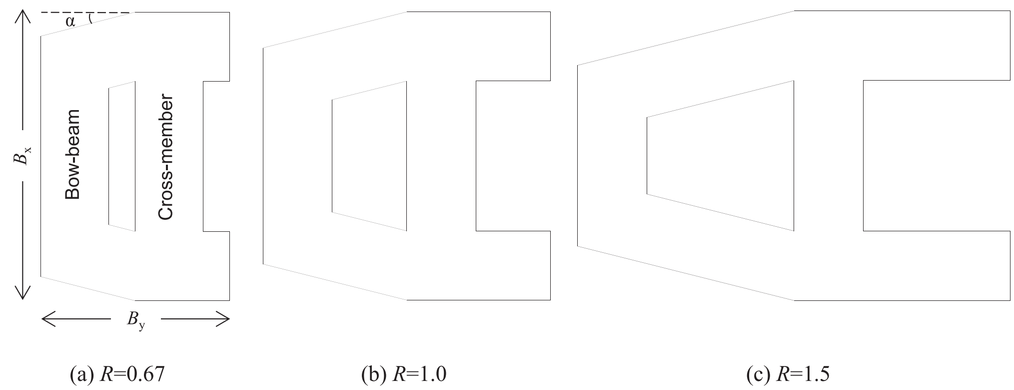


Figure 12. A-shaped mats with different values of *R*. (This figure is available in colour online.)

3.3. Comparison with the centrifuge test result

Taking the mat used in the centrifuge test as the research object, the ECM and the FEM result of vertical bearing capacity were calculated respectively, and compared with the test result.

The length of the idealised rectangular foundation along x-axis and y-axis (B_x', B_y') were calculated by solving Equations (1)–(3) according to the dimensions of the prototype mat. The calculations were divided into two cases: for action effects in x-direction and y-direction. The values of parameter used in computations are shown in Table 1.

The theoretical result of ultimate vertical load was solved according to Equation (4) with the idealised foundation dimensions (see Equation (5)).

$$V_{\text{ult}} = \begin{cases} 2337\ \text{kN} & (x\text{-direction}) \\ 2386\ \text{kN} & (y\text{-direction}) \end{cases} \tag{5}$$

The FE analysis of the vertical bearing capacity of the A-shaped mat was carried out according to the method in Section 3.2. The vertical load-displacement curve of the mat was obtained by controlling the displacement. The ultimate vertical load is 2524 kN, as shown in Figure 10.

The two ECM results and the FEM result of ultimate vertical load are compared with the centrifuge test value in Figure 11. It can be seen that the FEM result is in good agreement with the centrifuge test value, which verifies the reliability of FEM. The test result differs significantly from the two values of ECM. The ECM result is closely related to the selection of coordinate axis direction, because it involves the calculation of areal moment of inertia. This causes that for the same mat foundation, setting coordinate axes in different directions will get different vertical bearing capacity. It is obviously inconsistent with the real phenomenon. The ECM can be appropriately modified by using FEM results to avoid the problem of multivalued results and improve the accuracy of solutions.

3.4 Effect of the A-shaped mat dimensional parameters

Structure dimensional parameters such as the aspect ratio of shallow foundations have a significant influence on vertical bearing capacity. There are many geometric variables of the irregular A-shaped mats. The bow angle α and the width of each part of mat were kept as constants, B_x was also invariant. A series of mats with different aspect ratios were constructed by adjusting B_y (see Figure 12) to study the influence of aspect ratio and minimise the interference of other dimensional parameters. The aspect ratio $R = B_y/B_x$ ($B_x = 50$ m), and 4 kinds of A-shaped mats of which $R =$ 0.67, 1.0, 1.5, 2.0 were studied.

On this basis, B_x and B_y were invariant, a series of mats with different cross-member positions were constructed by adjusting the location of the cross-member in the y-direction for each aspect ratio of mats. Three typical types are shown in Figure 13. The cross-member position parameter J ($0 \leq J \leq 1$) was introduced, and its value indicates where the cross-member is located on the mat. When $J = 0$, it means that the cross-member is located at the stern end of the mat, as shown in Figure 13(a); when $J = 0.5$, it means that the cross-member is in the middle of the foundation, as shown in Figure 13(b); and when $J = 1$, it means that the cross-

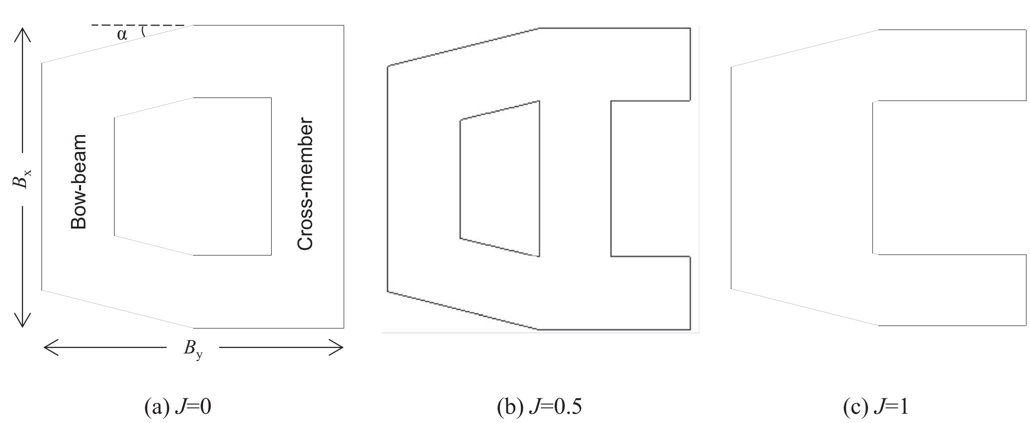


Figure 13. A-shaped mats with different values of *J*. (This figure is available in colour online.)

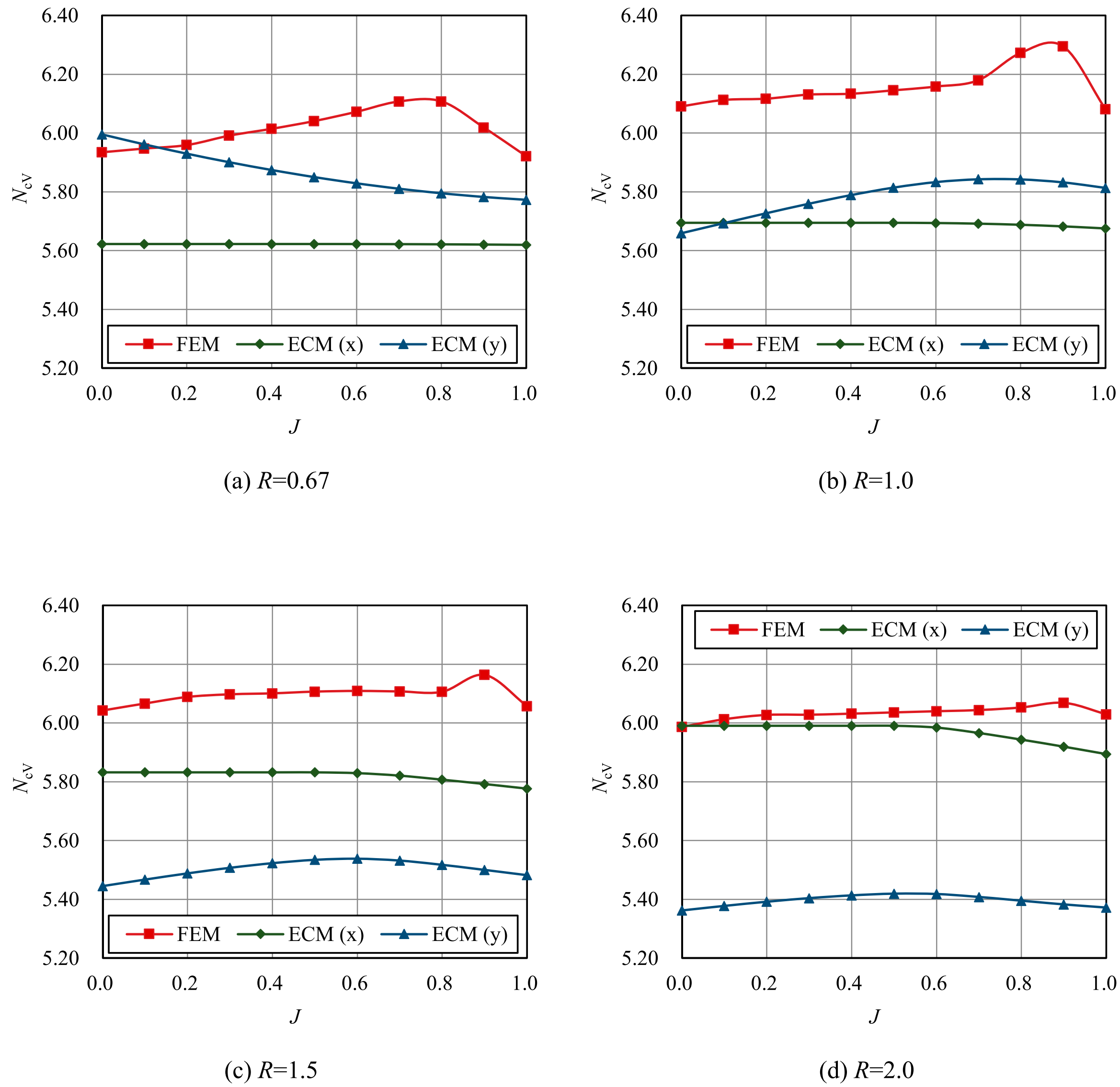


Figure 14. The variation of N_{cV} with J for different values of R. (This figure is available in colour online.)

member is located at the bow end of the mat and merges with the bow-beam, as shown in Figure 13(c).

Various mats may have different bottom areas due to the change of dimensional parameters. The bearing capacity factor was obtained by nondimensionalise the FEM result to eliminate the interference of bottom area on the analysis of bearing capacity. The vertical bearing capacity factor N_{cV} is calculated according to Equation (6).

$$N_{cV} = \frac{V_{ult}}{A\, s_u} \tag{6}$$

The FEM results and the two ECM results calculated by Equations (1)–(3) (for action effects in x-direction and y-direction) were converted to vertical bearing capacity factors. Figure 14 shows the variation curve of N_{cV} with the cross-member position parameter J for different A-shaped mats.

It can be seen from the FEM results in Figure 14 that the bearing capacity per unit area of the A-shaped mat increases gradually at first as the cross-member moves from the stern to the bow, and it decreases when the cross-member and the bow-beam are close enough or merged. This phenomenon can be explained in Figure 15. The perforation size of the mat bow reduces with the shortening of the distance between the cross-member and the bow-beam. The control capacity of mat on the soil between the two beams is enhanced. The plastic area of the soil is expanded, and the bearing performance is improved (see Figure 15(b)). The overlapping parts of the failure range of the soil (slip area) below each beam enlarge and merge into a single failure mechanism gradually when the two beams are too close. It makes the soil with obvious plastic deformation (dark area in the figure) becomes less, and the bearing capacity per unit area reduces after the coupling effect (see Figure 15(c)). At the same time, it is noted that the length ratios of long side to short side of the mat with $R = 0.67$ and $R = 1.5$ are both 1.5. However, the variation laws of N_{cV} are not the same, which reflects the asymmetric effect caused by the irregular shape of the mat. It is different from the rectangular foundation.

Correction calculations were carried out. As described in Section 3.3, the limitation of ECM makes it possible to have several theoretical solutions to the vertical bearing capacity. Only ECM results of the x-direction were discussed here for the convenience of modification. The shape correction factor f_{sV} is defined as Equation (7).

$$f_{sV} = N_{cV,F}/N_{cV,E} \tag{7}$$

where: $N_{cV,\,F}$ and $N_{cV,\,E}$ represent the FEM solution and the ECM solution of the vertical bearing capacity factor of mats, respectively.

The solid line in Figure 16 shows the law of f_{sV} changing with J and R. It indicates that f_{sV} generally shows an upward trend and declines at the end with the increase of J for the 4 kinds of A-shaped mats. The difference between the maximum and minimum values of the correction factor is small. The curve is appropriate for linear fitting, considering the simplicity of the correction process and the convenience of practical application. To avoid the over-estimation of the solution, the curve is fitted with piecewise function, as shown in Equations (8) and (9). The dotted line in Figure 16 represents the fitting result.

For $0 \leq J \leq 0.8$:

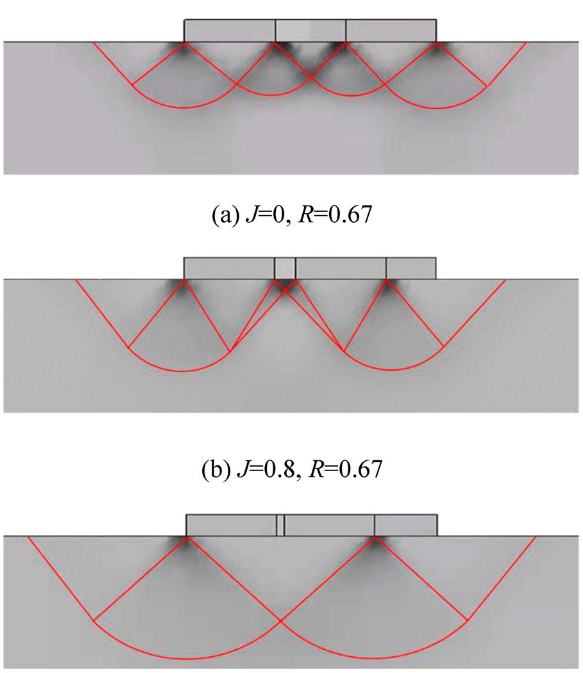
$$\begin{cases} f_{sV} = 0.02J + b & \\ b = 0.03R + 1.04 & (R \leq 1) \\ b = -0.07R + 1.14 & (R > 1) \end{cases} \tag{8}$$

For $0.8 < J \leq 1$:

$$\begin{cases} f_{sV} = kJ + b & \\ k = 0.12R - 0.19 & (R \leq 1) \\ k = 0.11R - 0.18 & (R > 1) \\ b = -0.07R + 1.21 & (R \leq 1) \\ b = -0.16R + 1.30 & (R > 1) \end{cases} \tag{9}$$

where: k is the slope, and b is the intercept of the fitting line.

The vertical bearing capacity of A-shaped mats with different dimensional parameters can be obtained by combining Equations (1)–(4), Equations (6)–(9) and the undrained shear strength of the soil.



(a) J=0, R=0.67

(b) J=0.8, R=0.67

(c) J=0.9, R=0.67

Figure 15. Soil failure mechanisms of mats with different values of J. (This figure is available in colour online.)

4. Study on bearing capacity of combined loading

The loads acting on mat foundations are multi-dimensional, including vertical, horizontal and moment loads, due to the combined action of environmental loads such as wind, wave and current and the dead-weight of platform structures. Failure envelopes of the foundation under combined loading should be analyzed. Probe tests were used to study bearing characteristics of mats in multi-dimensional load space.

4.1. Combined H-M capacity (V = 0)

4.1.1 Effect of R on H-M envelopes

The aspect ratio R affects the shape of H-M envelopes of the rectangular foundation, especially when the load acts along the major axis (Feng et al. 2017). Four types of mats with different values of R and constant J ($J = 0.5$) were analyzed by FEM to investigate the influence of R on the H-M envelope (see Figure 17). The horizontal and moment-bearing capacity factors N_{cH}, N_{cM} were computed by Equation (10).

$$\begin{cases} N_{cH} = \dfrac{H_{ult}}{A\, s_u} \\ N_{cM} = \dfrac{M_{ult}}{A\, B_y\, s_u} \end{cases} \tag{10}$$

where: H_{ult} and M_{ult} are ultimate horizontal and moment loads, respectively.

Figure 17(a) shows that the ability of the mat to resist pure moment load grows as R increases, but the growth rate decreases gradually. The ability to resist pure horizontal load rises firstly and then drops, and the maximum value is obtained when $R = 1$, but the difference is not obvious. The increase in R means that the length of the mat along the y-axis is getting longer and longer than the length along the x-axis, so the bearing capacity for the moment around the x-axis (which is more common in practice) will be higher. The theoretical value of N_{cH} is 1 for rectangular foundations on clay (API RP 2GEO 2014). The A-shaped mat has noticeable perforations, and the controlled soil area will be greater than the net bottom area of the foundation, so N_{cH} is greater than unity for the mat. At the same time, the subsoil will be pushed up to form a soil arch at the edge of the mat on horizontal movement direction in FE analysis (see Figure 18), which is also a factor not considered in the theoretical value. A smaller R represents a relatively long 'soil arch edge', making N_{cH} higher. However, the

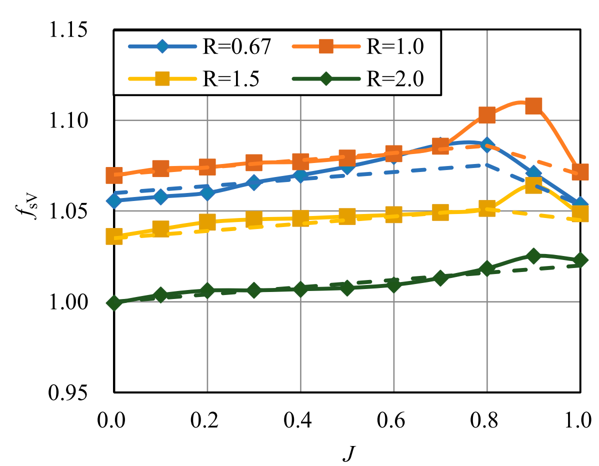


Figure 16. f_{sV} values under different J and R conditions. (This figure is available in colour online.)

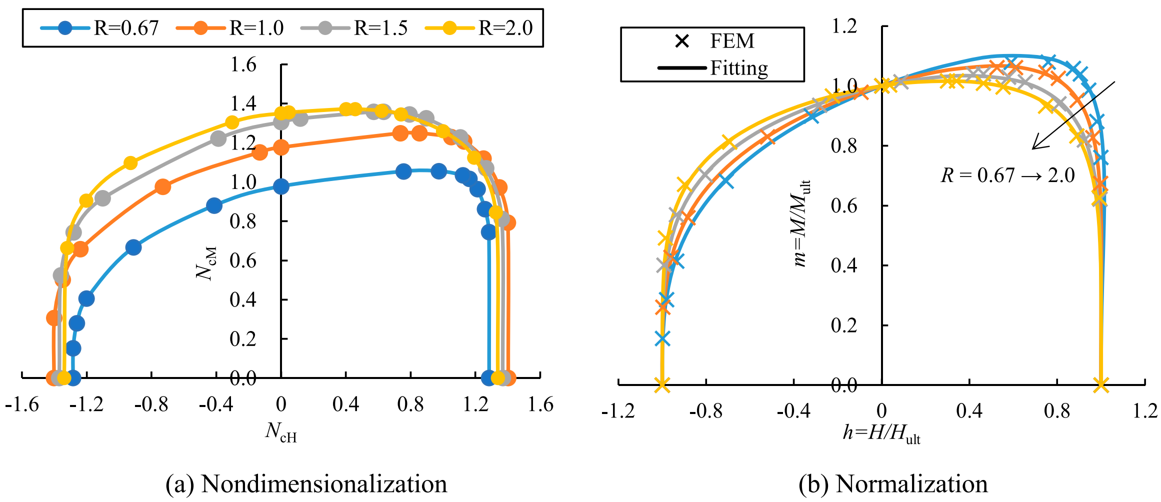


Figure 17. *H-M* envelopes of mats with different values of *R*. (This figure is available in colour online.)

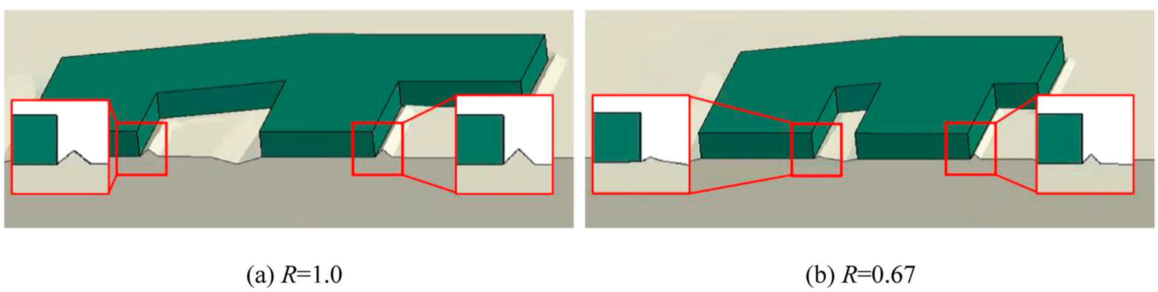


Figure 18. Soil arches formed by horizontal movement of mats. (This figure is available in colour online.)

cross-member is too close to the bow-beam to form an obvious soil arch when R is small enough (e.g. $R = 0.67$), which in turn lowers N_{cH} (see Figure 18(b)).

Figure 17(b) indicates that *H-M* envelopes of mats are asymmetrical. The envelope area of the first quadrant (H, M loads in the same direction) is generally greater than that of the second quadrant (H, M loads in the opposite direction). That is, the ability of the A-shaped mat to resist combined load of *H-M* can be relatively improved when subjected to the same direction H and M loads. The degree of asymmetry of the envelope rises with the decrease of R, and its law can be fitted by Equation (11).

$$h^2 + c\,h\,m^f + m^g = 1 \tag{11}$$

where: h and m are the normalised horizontal and moment loads, respectively ($h = H/H_{ult}$, $m = M/M_{ult}$); the coefficient c and the

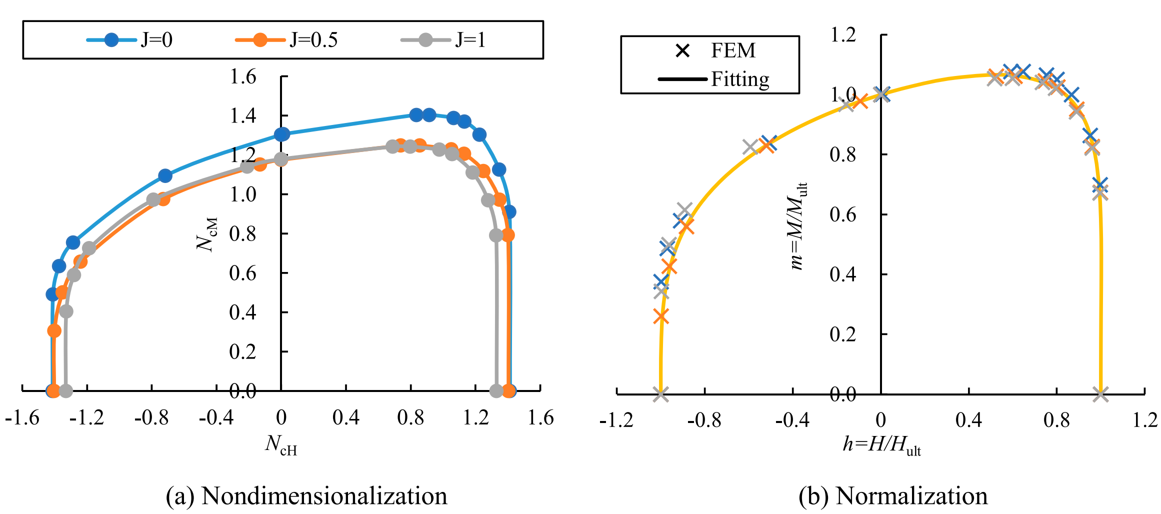


Figure 19. *H-M* envelopes of mats with different values of *J*. (This figure is available in colour online.)

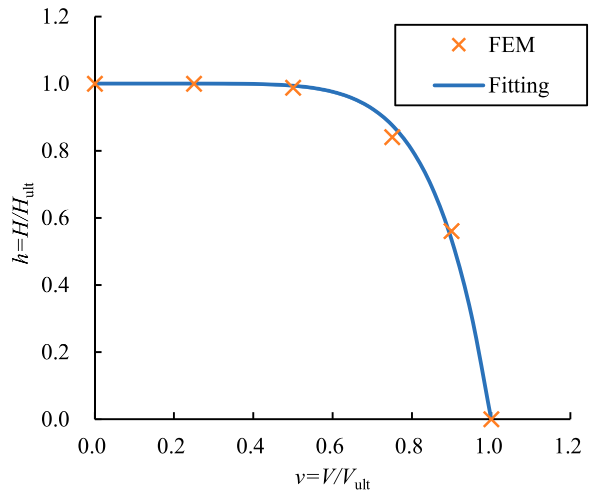


Figure 20. Normalised *V-H* envelope. (This figure is available in colour online.)

exponents *f*, *g* can be calculated by Equation (12).

$$\begin{cases} c = 0.30R - 1.13 \\ f = 1.37R + 2.06 \\ g = 1.14R + 2.61 \end{cases} \tag{12}$$

4.1.2 Effect of J on H-M envelopes

Three types of mats with different values of J ($J = 0, 0.5, 1$) and constant R ($R = 1$) were analyzed by FEM to investigate the influence of the cross-member position on the *H-M* envelope (see Figure 19).

The capacity to resist pure horizontal and moment loads of the mat grows as J decreases (i.e. the cross-member moves from the bow to the stern), and the envelope expands outward gradually. However, the changing J has little effect on *H-M* envelopes compared with the change of R. Figure 19(b) shows that the normalised *H-M* envelopes under different J conditions are basically the same shape and can still be obtained by Equations (11) and (12).

4.2. Effect of V on combined H-M capacity

The study of vertical bearing capacity and *H-M* envelope alone cannot completely evaluate the bearing safety of A-shaped mats. It is also necessary to analyze the impact of vertical load on combined *H-M* capacity, i.e. the three-dimensional *V-H-M* failure envelope.

It is necessary to clarify the range of h and m ultimate values under different v actions for deducing the mathematical expression of *H-M* envelopes under different levels of vertical loads. Therefore, the normalised *V-H* and *V-M* envelopes were investigated by FEM for the typical A-shaped mat of $R = 1$ and $J = 0.5$, as shown in Figures 20 and 21.

The ability to resist horizontal and moment loads of the mat reduces with an increasing vertical load. The envelopes coincide with the power function, which can be expressed by Equation (13).

$$\begin{cases} h = 1 - v^p \\ m = 1 - v^q \end{cases} \tag{13}$$

where: the exponents p and q are equal to 7.26 and 2.71, respectively.

Further, the normalised *H-M* envelopes when $v = 0.25, 0.5, 0.75, 0.9$ were computed, respectively. The envelopes were plotted together with the result of $v = 0$ in Figure 22 (expressed as cross dots).

The *H-M* envelopes contract inward gradually as v rises. The cost of bearing more vertical load is to sacrifice part of the combined *H-M* capacity for mats. The *H-M* envelope accords with the law in Section 4.1 when v is small. It has obvious asymmetry and has a higher combined bearing performance in the case of H and M loads in the same direction (the first quadrant). While the *H-M* envelope shape becomes more symmetrical when subjected to large vertical loads (e.g. $v = 0.75$). The envelope is difficult to fit through a single form of expression under different v conditions. The expression form needs to change with v.

For this purpose, the *V-H-M* envelope expression is constructed in the form of Equation (14), referring to Equations (11) and (13).

$$\left(\frac{h}{1-v^p}\right)^2 + c\left(\frac{h}{1-v^p}\right)\left(\frac{m}{1-v^q}\right)^f + \left(\frac{m}{1-v^q}\right)^g = 1 \tag{14}$$

where: the coefficient c and the exponents f, g coincide with the quadratic functions of v, as shown in Equation (15).

$$\begin{cases} c = 1.39v^2 - 0.09v - 0.84 \\ f = 0.49v^2 - 1.96v + 3.43 \\ g = 1.97v^2 - 3.10v + 3.76 \end{cases} \tag{15}$$

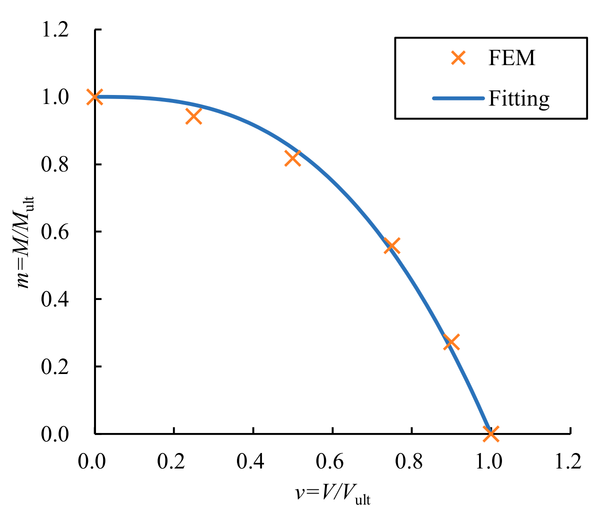


Figure 21. Normalised *V-M* envelope. (This figure is available in colour online.)

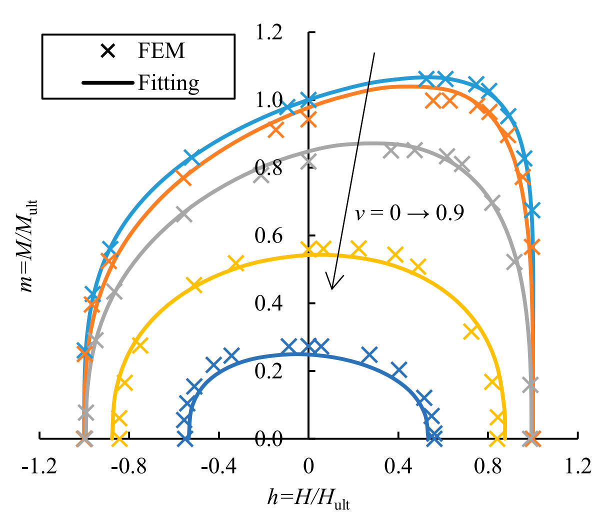


Figure 22. *H-M* envelopes under different *v* conditions. (This figure is available in colour online.)

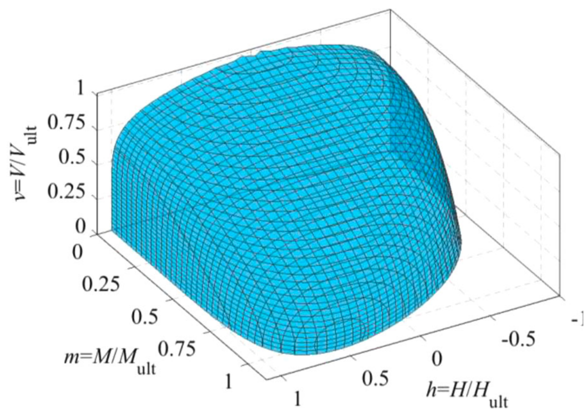


Figure 23. *V-H-M* failure envelope. (This figure is available in colour online.)

The solid lines in Figure 22 are the fitting curves expressed by Equation (14), which are in good agreement with FEM results. The three-dimensional *V-H-M* failure envelope of the A-shaped mat is shown in Figure 23.

5. Conclusions

This paper investigates the multi-dimensional bearing characteristics of A-shaped mat foundations on clay, and the centrifuge test was carried out for the vertical bearing capacity. Effects of dimensional parameters on uniaxial and multi-dimensional bearing capacities were analyzed. The expressions of the corresponding envelopes were constructed. The main conclusions are as follows:

(1) The vertical bearing capacity calculated by ECM is quite different from the centrifuge test result because of the irregular shape of the A-shaped mat. There is a multivalued result problem in ECM as well. A more accurate vertical bearing capacity compared with the test result can be better obtained after combining the modified method proposed in this paper.
(2) The horizontal bearing capacity of the A-shaped mat rises firstly and then reduces, and the moment bearing capacity continuously increases, but the growth rate drops gradually, with an increasing *R*. The horizontal and moment bearing capacity both decrease, and the *H-M* envelope contracts inward, as *J* rises. The *H-M* envelope is asymmetrical and the maximum value is obtained when *H* and *M* loads are in the same direction. The symmetry of *H-M* envelope increases with the growth of *R*, but it is not sensitive to *J*.
(3) The expressions of *H-M* envelopes of A-shaped mats under different dimensional parameters were constructed. Effects of vertical load on combined *H-M* capacity were analyzed. The *H-M* envelope area reduces, and its symmetry improves gradually, with the increase of vertical load on the mat. A new form of expression suitable for the three-dimensional *V-H-M* failure envelope of A-shaped mats was proposed.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The authors are grateful for the support provided by the National Science Foundation for Distinguished Young Scholars of China (grant number 51825904).

Data availability statement

The data used to support the findings of this study are included within the article.

References

API R P 2GEO. 2014. Geotechnical and foundation design considerations. Washington DC: *American Petroleum Institute*.
Boswell LF, Alekhin VN, Budarin AM. 2017. The jack-up platform for offshore oil and gas exploration overview. Russ J Constr Sci Technol. 3(2):5–12.
Butterfield R, Houlsby GT, Gottardi G. 1997. Standardized sign conventions and notation for generally loaded foundations. Géotechnique. 47(5):1051–1054.
Chen G, Liu R. 2018. Upper bound solutions of vertical bearing capacity of skirted mudmat in sand. Appl Ocean Res. 73:100–106.
Dassault Systèmes. 2011. Abaqus analysis users' manual. Providence (RI): Simulia Corp.
Davis E H, Booker J R. 1973. The effect of increasing strength with depth on the bearing capacity of clays. Géotechnique. 23(4):551–563.
DNV Classification Notes No. 30.4. 1992. Foundations. Høvik: Det Norske Veritas.
Fan D. 2014. Research on several key technologies of jack-up platform with oil storage mat [PhD thesis]. Ocean University of China.
Feng X, Gourvenec S, Shen Z. 2017. Shape effects on undrained capacity of mudmat foundations under multi-directional loading. Ocean Eng. 135:221–235.
Gourvenec S. 2007. Shape effects on the capacity of rectangular footings under general loading. Géotechnique. 57(8):637–646.
Graham J, Pinkney RB, Lew KV, Trainor PGS. 1982. Curve-fitting and laboratory data. Can Geotech J. 19(2):201–205.
Hansen JB. 1970. A revised and extended formula for bearing capacity. Dan Geotech Inst Bull. 28:5–11.
Hirst TJ, Steele JE, Remy ND, Scales RE. 1976. Performance of mat supported jack-up drilling rigs. The Eighth Annual Offshore Technology Conference, Houston, Texas.
Li S, Zhang Y, Li D, Khoa HDV, Adersen KH, Jiang Z, Liu J. 2021. Geotechnical analysis of a Mat foundation supporting a wellhead platform. J Geotech Geoenvironmental Eng. 147(11):04021121.
Liu M, Liu R, Wang L. 2021. Failure envelops analysis for square mudmat foundations on undrained clays under three-dimensional loading. Ships Offsh Struct. 16(sup2):77–84.
Liu R, Liu M, Tian Y, Wu X. 2019. Effect of perforations on the bearing capacity of shallow foundation on clay. Canadian Geotechnical Journal. 56(5):746–752.
Liu R, Zhang H, Chen G, Liu M. 2020. Study on vertical bearing capacity of solid and perforated mudmats in clay. Ocean Eng. 198:106973.
Meyerhof GG. 1953. The bearing capacity of foundations under eccentric and inclined loads. *Proceedings of the 3rd International Conference on SMFE*. 1. p. 440–445.
Parikh A, Shukla SJ, Gandhi SR. 2021. Reviews on mitigation techniques of offshore jack up hazard by FEM analysis. *Proc Indian Geotech Conf* 2019, *Lect Notes Civ Eng*. 134:741–755.
Prandtl L. 1921. Eindringungsfestigkeit und Festigkeit von Schneiden. *Zeit Angew Math Mech*. 1:15.
Shen Z, Feng X, Gourvenec S. 2017. Effect of interface condition on the undrained capacity of subsea mudmats under six-degree-of-freedom loading. Géotechnique. 67(4):338–349.
Taiebat H A, Carter J P. 2000. Numerical studies of the bearing capacity of shallow foundations on cohesive soil subjected to combined loading. Géotechnique. 50(4):409–418.
Young AG, House HF, Turner RD, Helfrich SC. 1982. Foundation performance of mat-supported jackup rigs in soft clays. J Pet Technol. 34(12):2958–2966.
Zhang Z, Wang Y. 2015. A numerical study on motion responses of a mat-support jack-up during positioning. *International Conference on Offshore Mechanics and Arctic Engineering*, American Society of Mechanical Engineers.